

Software Project Management Plan

Simulated E-Banking System (SEBS)

1. Overview

1.1 Project Summary

1.1.1 Purpose, Scope, and Objectives

The purpose of this project is to design, develop, test, and deliver a Simulated E-Banking System (SEBS). The SEBS will simulate the functionality of an online banking platform in a safe testing environment that uses mock data and simulated transactions.

The scope of the project includes the development of a web-based application that allows users to perform typical banking actions such as logging in, viewing accounts, checking balances, transferring funds between accounts, and reviewing transaction history. The system will also provide administrative tools for monitoring activity and managing test accounts.

The objectives of the project are to:

- Deliver a functional simulated online banking platform.
- Provide secure authentication and account access functionality.
- Simulate common banking workflows including deposits, withdrawals, transfers, and account management.
- Demonstrate secure system design practices such as access levels and session management.

The SEBS system will not connect to real banking networks, payment systems, or financial institutions. All data used in the system will be simulated.

The official system requirements are described in the System Requirements Specification (SRS) for the Simulated E-Banking System.

1.1.2 Assumptions and Constraints

The following assumptions and constraints apply to the project.

Assumptions:

- The project will be developed for educational or demonstration purposes.
- The system will rely on simulated account data and mock transactions.
- Users will access the system through standard web browsers.
- The software will be locally hosted and accessible through a web browser on the host machine for demonstration purposes.

Constraints:

- The system must not interact with real financial networks or banking institutions.
- Development will use standard web technologies such as Python, Django, and html..
- The project must be completed within the defined project schedule and available resources.

1.1.3 Project Deliverables

The following deliverables will be produced during the project.

- Software Project Management Plan (SPMP)
- System Requirements Specification (SRS)
- System design document (SDD)
- Source code for the SEBS system
- Test plans and testing documentation
- Final deployed simulated e-banking application
- Final project report

All deliverables will be provided in digital form and stored in a version-controlled project repository on github.

1.1.4 Schedule and Budget Summary

The project schedule will consist of the following major phases:

1. Project planning and requirements definition
2. System architecture and design
3. Implementation of frontend and backend components
4. Integration and system testing
5. Documentation and final delivery

Each phase will include milestone checkpoints to verify progress and deliverables.

The budget for this project is none as all tools are free.

1.2 Evolution of the SPMP

This SPMP is considered a living document. Updates may be made during the development process as project requirements, schedules, or resources change.

Any revisions to the SPMP will be reviewed by the project team and recorded in version control. Updated versions will be distributed to all relevant project stakeholders, in this case in later submissions.

2. References

The following references are used in this project plan.

- IEEE Std 1058-1998 – IEEE Standard for Software Project Management Plans
- Simulated E-Banking System Software Requirements Specification (SRS)

3. Definitions

SEBS – Simulated E-Banking System

SRS – Software Requirements Specification

MFA – Multi-Factor Authentication

4. Project Organization

4.1 External Interfaces

The project interacts with the following external organizational entities:

- The client or acquiring organization requesting the simulated banking system, in this case the professor
- The development team responsible for building the software, our group

Communication between these entities will occur through submissions of documentation throughout project progress.

4.2 Internal Structure

The project development team will include the following roles:

- Group leader
- Software developers
- Documentation

The group leader coordinates communication between team members and the stakeholders and ensures progress toward project milestones.

4.3 Roles and Responsibilities

Group leader

- Oversees project schedule and deliverables
- Coordinates team communication
- Manages project progress

Software Developers

- Implement system functionality
- Develop frontend and backend components
- Perform integration tasks

Documentation Team

- Produce technical documentation
- Maintain planning and system documentation

5. Managerial Process Plans

5.1 Project Start-Up Plan

5.1.1 Estimation Plan

Project cost will be negligible as all tools needed are free or available, effort, and schedule estimates will be developed based on the size and complexity of the system requirements. Estimation will be performed using historical experience with similar or similar locally hosted web demonstration projects and simplified effort estimation techniques.

Project progress will be reviewed periodically to update schedule and effort estimates.

5.1.2 Staffing Plan

The project team will consist of several members with the following skills:

- Web application development
- Backend server development
- Software testing and quality assurance

- Technical documentation

Personnel assignments may vary depending on project phase.

5.1.3 Resource Acquisition Plan

The project requires the following resources:

- Development environments
- Source code repository and version control system
- Web development tools and frameworks
- Local hosting hardware

All resources are already available or free development tools.

5.1.4 Project Staff Training Plan

Team members may receive training in the following areas if required:

- Secure web application development
- Database management and schema design
- Testing methodologies
- Version control and collaboration tools

Training will be conducted through online materials, documentation, and mentoring.

5.2 Work Plan

5.2.1 Work Activities

Major project activities include:

- Requirements analysis
- System design
- Frontend development
- Backend development
- Integration and testing
- Documentation preparation
- Final system deployment

Each activity produces specific deliverables and milestones.

5.2.2 Schedule Allocation

The project schedule includes the following milestones:

1. Completion of requirements specification
2. Completion of system design document
3. Implementation of core banking features
4. Integration of frontend and backend components
5. Completion of testing and bug fixes
6. Final system delivery

Progress will be monitored against these milestones.

5.2.3 Resource Allocation

Resources allocated to project activities include:

- Development personnel
- Software tools
- Testing environments
- Database infrastructure
- Documentation tools

Resources will be distributed according to the project timeline.

5.2.4 Budget Allocation

Budget considerations are negligible as all resources are free or available.

5.3 Control Plan

5.3.1 Requirements Control Plan

Changes to system requirements will be reviewed by the project team and approved before implementation. All requirement changes will be documented and tracked.

5.3.2 Schedule Control Plan

Project progress will be tracked against scheduled milestones. If delays occur, corrective actions such as task reassignment or timeline adjustment may be implemented.

5.3.3 Budget Control Plan

Budget and resource usage will be negligible as all resources are free or available.

5.3.4 Quality Control Plan

Quality assurance activities include:

- Code reviews
- Unit testing
- Integration testing
- Security testing
- System validation against requirements

5.3.5 Reporting Plan

Project status will be communicated through:

- Regular project meetings
- Documentation updates
- Written progress reports in the form of documentation submissions

5.3.6 Metrics Collection Plan

Metrics collected during development include:

- Number of completed features
- Number of detected defects
- Test coverage results
- Milestone completion status

5.4 Risk Management Plan

Potential project risks include:

- Security vulnerabilities in the web application
- Development delays due to technical challenges
- Incomplete requirements or scope changes
- Integration issues between frontend and backend systems

Risk mitigation strategies include early testing, regular reviews, and clear documentation.

5.5 Project Closeout Plan

At the completion of the project:

- All deliverables will be finalized and reviewed
- Project documentation will be archived
- Final reports and lessons learned will be recorded

6. Technical Process Plans

6.1 Process Model

The project will follow a structured software development life cycle consisting of requirements analysis, design, implementation, testing, and deployment.

Each stage produces deliverables that must be approved before proceeding to the next stage.

6.2 Methods, Tools, and Techniques

Development will use common web development technologies including:

- HTML, and the Django library for the frontend
- Python for backend logic
- Version control systems for code management

6.3 Infrastructure Plan

The development environment will include:

- Personal computers with development tools installed
- Version control repositories in the form of google cloud for documentation and github for code

These resources support development, testing, and deployment of the SEBS platform.

6.4 Product Acceptance Plan

The SEBS system will be accepted when:

- All functional requirements are implemented
- Security features such as authentication and access levels operate correctly
- Simulated banking workflows function as expected
- Testing confirms the system meets requirements

Acceptance testing may include demonstrations and functional testing

7. Supporting Process Plans

7.1 Configuration Management Plan

All source code and documentation will be stored in github as a version control system. Changes to the system will be tracked and documented to maintain version consistency.

7.2 Verification and Validation Plan

Verification and validation activities include:

- Unit testing of software modules
- Integration testing of system components
- System testing against requirements
- Security testing and validation of authentication mechanisms

7.3 Documentation Plan

Documentation produced for the project includes:

- Requirements documentation
- Design documentation
- User guides
- Administrator guides
- Testing documentation
- Final project report

7.4 Quality Assurance Plan

Quality assurance activities include code reviews, test case validation, and verification that development practices follow established standards.

7.5 Reviews and Audits Plan

Regular project reviews will be conducted to evaluate progress and quality. These reviews include:

- Requirements review
- Design review
- Code review
- Final system functions review

7.6 Problem Resolution Plan

Software defects and issues will be tracked through a problem reporting system using github issue tracking. Issues will be analyzed, prioritized, and resolved according to project priorities.

7.7 Subcontractor Management Plan

This project does not currently involve subcontractors. If subcontracted work becomes necessary, additional management procedures will be defined.

7.8 Process Improvement Plan

During development, project processes will be reviewed to identify areas for strategy improvement. Lessons learned will be documented to improve progress and future development efforts.

8. Additional Plans

Additional considerations for the SEBS project include:

- Security practices for handling simulated account data
- User training and documentation for system operation
- Maintenance procedures for future updates
- Data privacy practices even though all data is simulated

9. Plan Annexes

Supporting materials may include:

- Work breakdown structure diagrams
- Project schedules or Gantt charts
- System architecture diagrams

10. Plan Index

SEBS – Simulated E-Banking System
SPMP – Software Project Management Plan
SRS – Software Requirements Specification
MFA – Multi-Factor Authentication